

Course Code: CSA1606

Course Name: Data Warehouse and Data Mining

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1. COVARIANCE ANALYSIS IN HEALTH INSURANCE

Question

The following table shows the annual premium increase rate (x_i) and the number of claims per 100 policyholders (y_i) in a health insurance company.

Premium Increase (%) (x_i)	Claims per 100 Policyholders (y_i)
3.2	15
4.0	18
5.5	25
4.8	20

Determine whether premium increase and claim frequency have a positive or inverse relationship using the covariance formula.

Step 1: Calculate the Mean of X and Y

The mean (average) is calculated using:

$$\bar{x} = \frac{\sum x_i}{n}$$

where

- $n = 4$

Mean of Premium Increase (X)

$$\bar{x} = \frac{3.2+4.0+5.5+4.8}{4} = \frac{17.5}{4} = 4.375$$

Therefore,

$$\bar{x} = 4.375$$

Mean of Claims (Y)

$$\bar{y} = \frac{15+18+25+20}{4} = \frac{78}{4} = 19.5$$

Therefore,

$$\bar{y} = 19.5$$

Step 2: Calculate Deviations

Find the difference of every value from its mean.

x_i	y_i	$x_i - \bar{x}$	$y_i - \bar{y}$	Product
3.2	15	$3.2 - 4.375 = -1.175$	$15 - 19.5 = -4.5$	5.2875

x_i	y_i	$x_i - \bar{x}$	$y_i - \bar{y}$	Product
4.0	18	-0.375	-1.5	0.5625
5.5	25	1.125	5.5	6.1875
4.8	20	0.425	0.5	0.2125

Now add the products.

$$5.2875 + 0.5625 + 6.1875 + 0.2125 = 12.25$$

Step 3: Calculate Covariance

Population covariance formula:

$$Cov(X, Y) = \frac{\sum(x_i - \bar{x})(y_i - \bar{y})}{n} = \frac{12.25}{4} = 3.0625$$

Hence,

$$Cov(X, Y) = 3.0625$$

If sample covariance is required,

$$Cov(X, Y) = \frac{12.25}{3} = 4.0833$$

Interpretation

- Covariance $> 0 \rightarrow$ Positive relationship
- Covariance $< 0 \rightarrow$ Negative (Inverse) relationship
- Covariance $= 0 \rightarrow$ No linear relationship

Since

$$3.0625 > 0$$

there is a **positive relationship**.

Conclusion

As the number of insurance claims increases, the company generally increases the annual premium rate. Thus, premium increase and claim frequency are **positively related**.

2. DATA SMOOTHING USING EQUAL-FREQUENCY (EQUAL-DEPTH) BINNING

Question

Patient systolic blood pressure readings are:

110, 112, 115, 115, 118, 120, 120, 122, 124, 124, 126, 126, 126, 126, 130, 135, 135, 138, 138, 138, 138, 140, 145, 150, 152, 160, 180

Perform local smoothing using equal-frequency (equal-depth) binning.

Step 1: Count Observations

There are

27

observations.

We divide them into

3

equal bins.

Each bin contains

$27/3 = 9$

values.

Bin 1

110, 112, 115, 115, 118, 120, 120, 122, 124

Mean

$$= \frac{110+112+115+115+118+120+120+122+124}{9} = \frac{1056}{9} = 117.33$$

Replace every value with 117.33

Smoothed Bin:

117.33, 117.33, 117.33, 117.33, 117.33, 117.33, 117.33, 117.33, 117.33

Bin 2

124, 126, 126, 126, 126, 130, 135, 135, 138

Mean

$$= \frac{1166}{9} = 129.56$$

Replace every value with 129.56.

Smoothed Bin:

129.56 repeated nine times.

Bin 3

138, 138, 138, 140, 145, 150, 152, 160, 180

Mean

$$= \frac{1341}{9} = 149$$

Replace every value with 149.

Smoothed Bin:

149 repeated nine times.

Final Smoothed Dataset

117.33, 117.33, 117.33, 117.33, 117.33, 117.33, 117.33, 117.33, 117.33,
129.56, 129.56, 129.56, 129.56, 129.56, 129.56, 129.56, 129.56, 129.56,
149, 149, 149, 149, 149, 149, 149, 149, 149

Conclusion

Equal-frequency binning reduces the effect of small fluctuations in blood pressure while preserving the overall trend in the dataset.

3. EQUAL-WIDTH BINNING FOR BMI RISK CATEGORIZATION

Question

BMI values:

18.5, 19.2, 21.8, 24.9, 25.5, 26.2, 26.4, 27.8, 29.2, 30.2, 30.4, 31.9, 32.4, 33.1, 33.6, 34.7, 38.2, 39.5

Perform **equal-width binning** and smooth using the **nearest bin boundary**.

Step 1: Calculate Range

Minimum BMI

18.5

Maximum BMI

39.5

Range

$$39.5 - 18.5 = 21$$

Step 2: Calculate Bin Width

Using 3 bins,

$$Width = \frac{21}{3} = 7$$

Therefore bins are

- Bin 1 = 18.5 – 25.5
- Bin 2 = 25.5 – 32.5
- Bin 3 = 32.5 – 39.5

Step 3: Smooth by Nearest Boundary

Bin 1

18.5, 19.2, 21.8, 24.9, 25.5

Nearest boundaries

18.5, 18.5, 18.5, 25.5, 25.5

Bin 2

26.2, 26.4, 27.8, 29.2, 30.2, 30.4, 31.9, 32.4

Nearest boundary values

25.5, 25.5, 25.5, 32.5, 32.5, 32.5, 32.5, 32.5

Bin 3

33.1, 33.6, 34.7, 38.2, 39.5

Nearest boundaries

32.5, 32.5, 32.5, 39.5, 39.5

Final Smoothed BMI

18.5, 18.5, 18.5, 25.5, 25.5,

25.5, 25.5, 25.5, 32.5, 32.5, 32.5, 32.5, 32.5,

32.5, 32.5, 32.5, 39.5, 39.5

Conclusion

Equal-width binning groups BMI values into equal numerical intervals and replaces each value with its nearest boundary, simplifying risk categorization.

4. DATA REDUCTION USING BIN MEDIANS (MEDICAL SUPPLY COSTS)

Question

Costs (₹):

120, 150, 160, 160, 140, 145, 145, 148, 150, 150, 155, 155, 155, 155, 165, 170, 170, 175,
175, 175, 175, 180, 190, 200, 210, 230, 300

Perform **bin median smoothing**.

Step 1: Sort the Data

120, 140, 145, 145, 148, 150, 150, 150, 155,
155, 155, 155, 160, 160, 165, 170, 170, 175,
175, 175, 175, 180, 190, 200, 210, 230, 300

Total observations = 27

Divide into 3 bins.

Each bin contains 9 observations.

Bin 1

120, 140, 145, 145, **148**, 150, 150, 150, 155

Median = 148

Replace all values by 148.

Bin 2

155, 155, 155, 160, **160**, 165, 170, 170, 175

Median = 160

Replace all values by 160.

Bin 3

175, 175, 175, 180, **190**, 200, 210, 230, 300

Median = 190

Replace all values by 190.

Final Smoothed Data

148, 148, 148, 148, 148, 148, 148, 148, 148,

160, 160, 160, 160, 160, 160, 160, 160, 160,

190, 190, 190, 190, 190, 190, 190, 190, 190

Conclusion

Replacing values with the median reduces noise while maintaining the central tendency of each bin.

5. STANDARDIZATION OF PATIENT AGE DATA

Question

Ages:

18, 22, 25, 42, 28, 43, 33, 35, 56, 28

Perform:

- (a) Mean Absolute Deviation (MAD)
- (b) Z-score for the first four observations

(a) Mean Absolute Deviation (MAD)

Step 1: Calculate Mean

$$\bar{x} = \frac{18+22+25+42+28+43+33+35+56+28}{10} = \frac{330}{10} = 33$$

Step 2: Calculate Absolute Deviations

Age	Deviation	Absolute Deviation
18	-15	15
22	-11	11
25	-8	8
42	9	9
28	-5	5
43	10	10
33	0	0
35	2	2
56	23	23

Sum

$$15 + 11 + 8 + 9 + 5 + 10 + 0 + 2 + 23 + 5 = 88$$

MAD

$$= \frac{88}{10} = 8.8$$

Therefore,

$$MAD = 8.8$$

(b) Compute Standard Deviation

Squared deviations

Age	$(x - \bar{x})^2$
18	225
22	121
25	64
42	81
28	25
43	100
33	0
35	4
56	529
28	25

Total

$$225 + 121 + 64 + 81 + 25 + 100 + 0 + 4 + 529 + 25 = 1174$$

Variance

$$= \frac{1174}{10} = 117.4$$

Standard deviation

$$= \sqrt{117.4} = 10.84$$

Step 3: Calculate Z-score

Formula

$$Z = \frac{x - \bar{x}}{\sigma}$$

For Age = 18

$$Z = \frac{18 - 33}{10.84} = \frac{-15}{10.84} = -1.38$$

For Age = 22

$$z = \frac{22-33}{10.84} = -1.01$$

For Age = 25

$$z = \frac{25-33}{10.84} = -0.74$$

For Age = 42

$$z = \frac{42-33}{10.84} = 0.83$$